

Unit-4 : Security, Privacy, and Fraud Detection in OSNs

Dr.Pooja Bhatt

Privacy and Security in Online Social Media (03M10002UE01)
Artificial Intelligence and Data Science Department

Contents

1. Identity Management Overview.....	3
2. Key Components of Identity Management in OSNs...	4
3. Types of Identities in OSNs.....	5
4. Identity Management Models in OSNs.....	6
5. Challenges in Identity Management in OSNs.....	7
6. Future Trends in Identity Management.....	8
7. Case Study: Facebook Identity Management & Privacy Controversies	9
8. digital identity.....	10
9. Key Components of Digital Identity.....	11
10. Types of Digital Identity.....	12
11. Digital Identity Lifecycle.....	13
12. Digital Identity Management Models.....	14
13. Identity Verification Methods in Online Social Networks (OSNs)	15

Identity Management Overview

Identity management (IdM) in **online social networks (OSNs)** involves the **creation, authentication, verification, and maintenance of user identities** to ensure security, privacy, and controlled access to personal information. With billions of users interacting on platforms like Facebook, Instagram, and Twitter, identity management plays a crucial role in **preventing identity theft, fraud, and unauthorized access**.

Effective identity management ensures:

- **Secure user authentication** (passwords, 2FA, biometrics).
- **Privacy protection** by allowing users to control visibility.
- **Prevention of fake accounts, bots, and identity fraud**.
- **Trust and credibility** in online interactions.

With the rise of **AI-driven identity verification, blockchain-based decentralized identities, and stronger regulations**, the future of identity management in OSNs is evolving toward **enhanced security and user autonomy**.

Key Components of Identity Management in OSNs

Component	Description	Example in OSNs
User Registration	Users create accounts using an email, phone number, or social login.	Facebook, LinkedIn
Authentication	Validating a user's identity using passwords, two-factor authentication (2FA), or biometrics.	Google 2FA, Instagram's login alerts
Profile Management	Users customize their profile, control visibility, and manage personal details.	Twitter private accounts, Facebook profile privacy
Access Control & Permissions	Determines who can view, comment, and interact with a user's content.	LinkedIn profile visibility settings
Identity Verification	Verifies users through phone verification, ID upload, or AI-powered face recognition.	Twitter (X) verified checkmarks, Instagram identity verification
Fraud Detection & Prevention	Detects and blocks fake profiles, bots, and impersonators using AI	Facebook AI-based fake account removal

Types of Identities in OSNs

Identity Type	Key Feature	Risk Level
Real	Authentic & verifiable	Low
Pseudonymous	Consistent alias	Medium
Anonymous	No traceable data	High
Fake	Deceptive	Very High
Bot	Automated	Medium–High

Identity Management Models in OSNs

Model	Control	Privacy	Scalability	Common Use
Centralized	Platform	Low	High	Mainstream OSNs
Federated	Shared	Medium	High	Cross-platform login
User-Centric	User	High	Medium	Privacy-focused apps
Decentralized (SSI)	User	Very High	Low–Medium	Web3, research
Attribute-Based	Shared	High	Medium	Privacy-sensitive access
Role-Based	Platform	Low	High	Community management

Challenges in Identity Management in OSNs

- Fake Profiles & Impersonation – Fraudulent users create fake accounts to spread misinformation, scam others, or manipulate public opinion.
- Identity Theft & Phishing Attacks – Cybercriminals steal user credentials to access private data or commit fraud.
- Data Privacy Issues – OSNs collect vast amounts of personal data, increasing privacy concerns and risks of data breaches.
- Multiple Account Handling – Managing different identities across multiple platforms is complex.
- Regulatory Compliance – OSNs must adhere to strict privacy laws (GDPR, CCPA, Digital Services Act) while balancing business interests.

Future Trends in Identity Management

- **AI-Driven Identity Verification** – OSNs will use AI and machine learning to detect fake profiles, deep fakes, and bot activity.
- **Biometric Authentication** – Platforms will increasingly adopt fingerprint, facial recognition, and voice authentication for secure logins.
- **Self-Sovereign Identity (SSI)** – Blockchain-based identity solutions will give users full control over their personal data.
- **Zero-Knowledge Proofs (ZKP)** – Users will verify their identity without revealing sensitive details, enhancing privacy.
- **Decentralized Identifiers (DIDs)** – Users will store their digital identities on blockchain networks rather than relying on centralized social media companies.

Case Study: Facebook Identity Management & Privacy Controversies

Facebook, as one of the largest OSNs, has faced multiple identity management challenges, including:

-  2018 Cambridge Analytica Scandal – Millions of user identities were compromised for political ad targeting.
-  Fake Account Epidemic – Facebook removed 5.5 billion fake accounts in 2023 to combat impersonation and misinformation.
-  Facebook Login Exploits – Security breaches exposed personal details of users who used Facebook to log into third-party apps.

To address these issues, Facebook introduced:

- ✓ Identity Verification for Blue Checkmarks – Ensures accounts of public figures and businesses are authentic.
- ✓ Stronger Privacy Controls – Users can control who sees their posts, profile info, and personal data sharing.
- ✓ AI-Based Fake Account Detection – Uses machine learning to detect and remove bot accounts and impersonators.

Digital Identity – Introduction

- A digital identity is a unique online representation of an individual, organization, or device that allows access to digital systems and services.
- It includes personal data, authentication credentials, and behavioral information used for verification, security, and access control.
- Digital identities are crucial in various domains, including online social networks (OSNs), banking, e-commerce, healthcare, and government services.

Why is Digital Identity Important?

- ✓ Enables secure authentication for online interactions.
- ✓ Protects users from identity theft and fraud.
- ✓ Allows for personalization of online services.
- ✓ Ensures privacy and data protection.
- ✓ Facilitates compliance with regulatory frameworks like GDPR, CCPA, and HIPAA.

Key Components of Digital Identity

Component of Digital Identity	Description
Identifiers	Unique elements that distinguish one digital entity from another, such as user ID, username, email ID, or phone number.
Attributes	Personal or profile-related information that describes the user; may be static (name, DOB) or dynamic (location, status).
Credentials	Secret or secure data used to prove ownership of a digital identity, such as passwords, PINs, or cryptographic keys.
Authentication Mechanisms	Processes and techniques used to verify a user's identity, including password-based, OTP, biometric, or multi-factor authentication.
Authorization & Access Rights	Defines what resources or actions a user is permitted to access after successful authentication.
Reputation & Trust Metrics	Indicators of credibility and reliability of a digital identity based on past behavior and social interactions.
Privacy Preferences & Consent	User-defined settings that control visibility, data sharing, and permissions related to personal information.
Digital Footprints	Records of online activities left by users, including posts, likes, comments, searches, and interaction history.

Types of Digital Identity

Identity Type	Description	Examples
Personal Digital Identity	Used by individuals for personal online interactions and social networking activities.	Facebook profile, Google account, Twitter handle
Corporate Digital Identity	Used by organizations to authenticate employees, partners, and customers in digital systems.	LinkedIn business page, corporate email accounts
Government-Issued Digital Identity	Official digital identity issued by government authorities to access public services securely.	Aadhaar (India), e-Estonia, US Social Security Number (SSN)
Decentralized Digital Identity (DDI)	User-controlled digital identity managed without a central authority, often using blockchain technology.	Blockchain-based IDs, Self-Sovereign Identity (SSI)
Anonymous Identity	Allows users to interact online without revealing personal or identifiable information.	Tor identities, dark web accounts, 4chan, Whisper

Digital Identity Lifecycle

A digital identity undergoes various stages throughout its existence.

- 1 Identity Creation – A user registers an account using personal details.
- 2 Authentication – The system verifies the user through credentials (password, OTP, biometrics).
- 3 Authorization & Access Control – The user gets access based on roles and permissions.
- 4 Identity Usage – The user engages in online activities (e.g., social media posting, transactions).
- 5 Monitoring & Identity Management – Continuous tracking to detect suspicious activities.
- 6 Identity Deactivation or Deletion – The identity is revoked or deleted upon user request or policy enforcement.

Digital Identity Management Models

Model	Description	Examples	Advantages	Disadvantages
Centralized Identity Management	A single provider controls the identity database.	Facebook Login, Google Sign-In	Simple, widely used, easy integration	Privacy risks, prone to data breaches
Federated Identity Management (FIM)	Users use a single identity for multiple services.	Google & Microsoft accounts for third-party logins	Reduces password fatigue, seamless user experience	If the central account is compromised, multiple accounts are affected
Decentralized Identity Management (DDI)	Users manage their identity without third parties, often using blockchain.	Self-Sovereign Identity (SSI), Web3-based logins	Enhanced privacy, greater user control	Complex implementation, limited adoption

Identity Verification Methods in Online Social Networks (OSNs)

Method	Description	Example in Social Media
Username & Password	Traditional method of authentication.	Logging into Facebook with a password
Two-Factor Authentication (2FA)	Requires an additional verification step.	Google sending an OTP to a registered phone
Biometric Authentication	Uses fingerprints, face, or voice recognition.	Face ID login on Apple devices
Behavioral Analysis	Tracks typing speed, mouse movements, and login patterns.	AI detecting unusual login attempts on Twitter
AI-Powered Verification	AI checks uploaded documents or selfies for identity proof.	Instagram verifying users through government-issued ID

Solutions & Future Trends in Digital Identity

- ✅ **Biometric Authentication** – Increasing use of fingerprints, facial recognition, and voice ID.
- ✅ **Self-Sovereign Identity (SSI)** – Blockchain-based digital IDs for full user control.
- ✅ **Zero-Knowledge Proofs (ZKP)** – Allows verification without exposing personal data.
- ✅ **AI & Machine Learning for Fraud Detection** – AI-based anomaly detection to prevent fake accounts and fraud.
- ✅ **Multi-Factor Authentication (MFA)** – Combining password, OTP, biometrics for stronger security.
- ✅ **Decentralized Identity Networks** – Platforms like Hyperledger Indy and Microsoft ION using blockchain for secure identities.

Case Study: Digital Identity in Action

Case Study 1: Facebook & Cambridge Analytica Scandal (2018)

- Issue: Personal data of 87 million users was collected without consent.
- Impact: Raised concerns about privacy violations in OSNs.
- Solution: Introduction of stricter privacy laws like GDPR.

Case Study 2: Twitter (X) Blue Checkmark Verification

- Issue: Verification was previously limited to notable users but changed to a paid model.
- Impact: Increased fake accounts and impersonation.
- Solution: AI-powered verification and stricter identity checks.

Case Study 3: India's Aadhaar Digital Identity System

- Issue: Largest biometric ID system with over 1.3 billion users.
- Impact: Facilitates e-Governance but raises privacy concerns.
- Solution: Implemented data protection measures to prevent misuse

Phishing Attacks in OSNs and Their Countermeasures

Aspect	Description
Meaning of Phishing in OSNs	Phishing attacks in Online Social Networks (OSNs) involve fake messages, links, or profiles designed to trick users into revealing passwords, OTPs, or financial information.
Common Types of Phishing Attacks	Fake login pages, malicious links in messages/comments, impersonation of trusted contacts, giveaway scams, and credential-harvesting pages.
Attack Techniques Used	Social engineering, URL spoofing, shortened links, fake verification badges, cloned profiles, and urgent or emotional messages.
Impact on Users & Platforms	Account takeover, identity theft, financial loss, data leakage, reduced user trust, and damage to platform reputation.
Detection Techniques	URL analysis, content filtering, behavioral analysis, anomaly detection, ML-based classifiers, and user-reported feedback.
Countermeasures & Prevention	User awareness training, two-factor authentication (2FA), AI-based phishing detection, link warnings, identity verification, and regular security updates.

Identifying Fraudulent Entities and Fake Profiles in OSNs.

Aspect	Description
Definition of Fraudulent Entities & Fake Profiles	Fake profiles are accounts created using false or stolen identities. Fraudulent entities engage in malicious activities such as phishing, scams, impersonation, and spreading misinformation on OSNs.
Common Characteristics	Newly created accounts, incomplete or copied profile information, abnormal friend/follower ratios, repetitive automated posts, and unusual login behavior.
Identification Techniques	Profile-based, behavior-based, network-based, content-based analysis, and AI/ML techniques are used to detect fraudulent accounts.
Role of AI & Machine Learning	AI analyzes large-scale OSN data to detect bots, coordinated fake behavior, suspicious patterns, and verify identity using images, documents, and biometrics.
Challenges in Detection	Rapidly evolving fraud strategies, high false positives, privacy issues, and difficulty detecting well-maintained fake profiles.
Importance & Impact	Protects users from scams, enhances platform trust, prevents misinformation, and ensures fair engagement and security in OSNs.

Parul[®]
University

NAAC
GRADE **A++**



<https://paruluniversity.ac.in/>

